

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

Managerial Economics

Section Id :	640653138799
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	10
Number of Questions to be attempted :	10
Section Marks :	45
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653386969
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532125059 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : MANAGERIAL ECONOMICS

(COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536925371.  Yes

6406536925372.  No

Sub-Section Number :

2

Sub-Section Id :

640653386970

Question Shuffling Allowed :

No

Question Id : 6406532125060 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (2 to 5)

Question Label : Comprehension

The market for paper in southern region in India is characterized by the following demand and supply curves: $Q_D = 160,000 - 2000P$ and $Q_S = 40,000 + 2000P$, where Q_D is the quantity demanded in 100-pound lots, Q_S is the quantity supplied in 100-pound lots, and P is the price per 100-pound lot. Currently there is no attempt to regulate the dumping of effluent into streams and rivers by paper mills. As a result, dumping is widespread. The *Marginal External Cost (MEC)* associated with the production of paper is given by the curve $MEC = 0.0006Q_S$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 2 Question Id : 6406532125061 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

Calculate the output of paper if it is produced under competitive conditions and no attempt is made to monitor or regulate the dumping of effluent. $Q = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

100000

Question Number : 3 Question Id : 6406532125062 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

Calculate the Price of paper if it is produced under competitive conditions and no attempt is made to monitor or regulate the dumping of effluent. $P =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Number : 4 Question Id : 6406532125063 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

Determine the socially efficient price of paper. $P =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

48.70 to 48.80

Question Number : 5 Question Id : 6406532125064 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

Determine the socially efficient quantity of paper. $Q =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

62500

Sub-Section Number :

3

Sub-Section Id :

640653386971

Question Shuffling Allowed :

No

Question Id : 6406532125065 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (6 to 15)

Question Label : Comprehension

Suppose two firms operate in a market and can produce at a constant average (and marginal) cost of $AC = MC = \$10$. Let Q_1 be the output of the first firm and Q_2 be the output of the second. Market demand is given by $Q_1 + Q_2 = 70 - P$. Assuming that the firms are competing as per the Cournot competition, find out the equilibrium quantity for each firm, equilibrium price and profit (π_1, π_2) of each firm.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 6 Question Id : 6406532125066 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

$Q_1^* = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

20

Question Number : 7 Question Id : 6406532125067 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

$Q_2^* = \underline{\hspace{2cm}}$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

20

Question Number : 8 Question Id : 6406532125068 Question Type : SA Keyboard Layout :

Inscript

Correct Marks : 1

Question Label : Short Answer Question

$P^* =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Number : 9 Question Id : 6406532125069 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

$\pi_1 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

400

Question Number : 10 Question Id : 6406532125070 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

$\pi_2 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

400

Question Number : 11 Question Id : 6406532125071 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Now suppose that the two firms are competing in the Stackelberg manner where one of the firms makes its output decision before the other. Suppose Firm 1 is the Stackelberg leader (i.e., makes its output decisions before Firm 2). Calculate the equilibrium quantity for the firms, equilibrium market price and profits (π_1 , π_2) of both the firms in equilibrium.

$Q_1^* =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Number : 12 Question Id : 6406532125072 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Now suppose that the two firms are competing in the Stackelberg manner where one of the firms makes its output decision before the other. Suppose Firm 1 is the Stackelberg leader (i.e., makes its output decisions before Firm 2). Calculate the equilibrium quantity for the firms, equilibrium market price and profits (π_1 , π_2) of both the firms in equilibrium.

$Q_2^* =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

15

Question Number : 13 Question Id : 6406532125073 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Now suppose that the two firms are competing in the Stackelberg manner where one of the firms makes its output decision before the other. Suppose Firm 1 is the Stackelberg leader (i.e., makes its output decisions before Firm 2). Calculate the equilibrium quantity for the firms, equilibrium

market price and profits (π_1 , π_2) of both the firms in equilibrium.

$P^* =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

25

Question Number : 14 Question Id : 6406532125074 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Now suppose that the two firms are competing in the Stackelberg manner where one of the firms makes its output decision before the other. Suppose Firm 1 is the Stackelberg leader (i.e., makes its output decisions before Firm 2). Calculate the equilibrium quantity for the firms, equilibrium market price and profits (π_1 , π_2) of both the firms in equilibrium.

$\pi_1 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

450

Question Number : 15 Question Id : 6406532125075 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Now suppose that the two firms are competing in the Stackelberg manner where one of the firms makes its output decision before the other. Suppose Firm 1 is the Stackelberg leader (i.e., makes its output decisions before Firm 2). Calculate the equilibrium quantity for the firms, equilibrium market price and profits (π_1 , π_2) of both the firms in equilibrium.

$\pi_2 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

225

Sub-Section Number : 4

Sub-Section Id : 640653386972

Question Shuffling Allowed : No

Question Id : 6406532125076 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (16 to 19)

Question Label : Comprehension

Consider two investment opportunities in new products: gourmet marmite and gourmet honey. People might either really like or really dislike the gourmet marmite, so it generates a payoff of \$0.5 with 30 percent probability, a payoff of \$1 million with 40 percent probability and a payoff of \$1.5 million with 30 percent probability. The gourmet honey is less likely to either really turn people off or really turn them on, so it has a payoff of \$0.5 million with 20 percent probability, \$1 million with 60 percent probability and \$1.5 million with 20 percent probability.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 16 Question Id : 6406532125077 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Expected value of first product is _____(in \$)

NOTE: Enter the exact answer without any extra space in the beginning or at the end.

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Set

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

1 million

1million

1000000

Question Number : 17 Question Id : 6406532125078 Question Type : SA Keyboard Layout :

Inscript**Correct Marks : 1**

Question Label : Short Answer Question

Expected value of second product is _____ (in \$)

NOTE: Enter the exact answer without any extra space in the beginning or at the end.**Response Type :** Alphanumeric**Evaluation Required For SA :** Yes**Show Word Count :** Yes**Answers Type :** Set**Answers Case Sensitive :** No**Text Areas :** PlainText**Possible Answers :**

1 million

1million

1000000

Question Number : 18 Question Id : 6406532125079 Question Type : MSQ**Correct Marks : 2 Max. Selectable Options : 0**

Question Label : Multiple Select Question

Choose the correct alternative(s)

Options :

6406536925389. ✖ The second product is riskier than the first product

6406536925390. ✔ Second product has lower variance than the first product

6406536925391. ✖ First product has a lower variance than the second product

6406536925392. ✔ The first product is riskier than the second product

Question Number : 19 Question Id : 6406532125080 Question Type : MSQ**Correct Marks : 2 Max. Selectable Options : 0**

Question Label : Multiple Select Question

A risk-loving decision maker has the utility function $U(I) = 25I^2$ where I denotes the payoff of the product.

Choose the correct alternative(s) for this risk-loving decision maker

Options :

6406536925393. ✔ Expected utility of investing in gourmet marmite is more than expected utility of investing in gourmet honey

6406536925394. ✖ A risk-loving decision maker prefers investing in gourmet honey to investing in gourmet marmite

6406536925395. ✖ Expected utility of investing in gourmet marmite is less than expected utility of investing in gourmet honey

6406536925396. ✔ A risk-loving decision-maker prefers investing in gourmet marmite to investing

in gourmet honey

Sub-Section Number : 5
Sub-Section Id : 640653386973
Question Shuffling Allowed : No

Question Id : 6406532125081 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (20 to 25)

Question Label : Comprehension

India currently imports all of its palm oil. The annual demand for palm oil by Indian consumers is given by the demand curve $Q=2000-4P$, where Q is quantity of palm oil in thousand litres and P is market price in INR per litre. World producers can harvest and ship palm oil to Indian distributors at a constant marginal cost (=average cost) of INR 50 and Indian distributors can distribute it at a cost of INR 10 per litre. Consider market for palm oil to be competitive.
Indian government is considering a tariff on palm oil import of INR 10 per litre.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 20 Question Id : 6406532125082 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

If there is no tariff, what is the quantity demanded (in thousand litres) _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1760

Question Number : 21 Question Id : 6406532125083 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

If the tariff is imposed, how much a consumer will pay for a litre of palm oil _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

Question Number : 22 Question Id : 6406532125084 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

What is the new quantity demanded now (in thousand litres)? _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1720

Question Number : 23 Question Id : 6406532125085 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

What will be the lost consumer surplus due to the imposed tariff (in thousand INR) _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

17400

Question Number : 24 Question Id : 6406532125086 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

Tax revenue for the government will be (in thousand INR) _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

17200

Question Number : 25 Question Id : 6406532125087 Question Type : MCQ

Correct Marks : 1.5

Question Label : Multiple Choice Question

Tariff results in

Options :

6406536925402. ✖ Net gain for the society

6406536925403. ✔ Net loss for the society

6406536925404. ✖ Neither loss nor gain

6406536925405. ✖ Cannot compute

Sub-Section Number :

6

Sub-Section Id :

640653386974

Question Shuffling Allowed :

No

Question Id : 6406532125088 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (26 to 29)

Question Label : Comprehension

A monopolist faces the demand curve $P = 48 - 6Q$, where P is measured in dollars per unit and Q in thousands of units. The monopolist has a constant average cost of \$12 per unit.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 26 Question Id : 6406532125089 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the monopolist's profit-maximizing price and quantity? $P^* =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

30

Question Number : 27 Question Id : 6406532125090 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the monopolist's profit-maximizing price and quantity? $Q^* =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3000

Question Number : 28 Question Id : 6406532125091 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

The resulting profit to the monopolist will be $\pi =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

54000

Question Number : 29 Question Id : 6406532125092 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is firm's degree of monopoly power using the Lerner index _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.6

Sub-Section Number : 7

Sub-Section Id : 640653386975

Question Shuffling Allowed : No

Question Id : 6406532125093 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (30 to 31)

Question Label : Comprehension

Mohan quits his IT job, where he was earning a salary of ₹700,000 per year, to start his own restaurant business in a building that he owns and was previously renting out for ₹290,000 per year. In his first year of business, he has the following expenses: salary paid to himself, ₹500,000;

rent, ₹0; other expenses, ₹340,000.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 30 Question Id : 6406532125094 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

The accounting cost associated with Mohan's restaurant business is _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

840000

Question Number : 31 Question Id : 6406532125095 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1.5

Question Label : Short Answer Question

The economic cost associated with Mohan's restaurant business is _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1330000

Sub-Section Number :

8

Sub-Section Id :

640653386976

Question Shuffling Allowed :

No

Question Id : 6406532125096 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (32 to 33)

Question Label : Comprehension

There are 40 low-risk people and 60 high-risk people. A low-risk person has an average of \$1500 in medical expenses each year and is willing to pay \$1800 for medical insurance (this person is risk-averse). A high-risk person has an average of \$2200 in medical expenses each year and is willing to pay \$2600 for medical insurance. Insurance companies are unable to differentiate who is high-risk

and who is low-risk.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 32 Question Id : 6406532125097 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the insurance provider offered medical insurance at a price of \$2000

Options :

6406536925412. ✖ Everyone will enroll for the insurance, irrespective of the type

6406536925413. ✔ The provider will lose money

6406536925414. ✖ Both Everyone will enroll for the insurance, irrespective of the type and The provider will lose money

6406536925415. ✖ None of these

Question Number : 33 Question Id : 6406532125098 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

If the insurance company offered medical insurance at a price of \$2400

Options :

6406536925416. ✔ Low-risk people would not be insured

6406536925417. ✖ Total surplus will be \$12000, if the price is \$2400

6406536925418. ✖ Both Low-risk people would not be insured and Total surplus will be \$12000, if the price is \$2400

6406536925419. ✖ None of these

Sub-Section Number :

9

Sub-Section Id :

640653386977

Question Shuffling Allowed :

No

Question Id : 6406532125099 Question Type : COMPREHENSION Sub Question Shuffling

Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (34 to 36)

Question Label : Comprehension

The production function for a product is given by $q = 450KL$. If the price of capital is INR 30 per day and the price of labour is INR 150 per day.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 34 Question Id : 6406532125100 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the cost minimizing quantity of labour and capital respectively for producing 9000 units of output? $L^* =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Question Number : 35 Question Id : 6406532125101 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the cost minimizing quantity of labour and capital respectively for producing 9000 units of output? $K^* =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

10

Question Number : 36 Question Id : 6406532125102 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the minimum cost of producing 9000 units of output in INR?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

600

Sub-Section Number :

10

Sub-Section Id :

640653386978

Question Shuffling Allowed :

No

Question Id : 6406532125103 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (37 to 38)

Question Label : Comprehension

Consider the utility function $U(X_1, X_2) = \log X_1 + \log X_2$ and the budget line of the consumer as $2X_1 + 6X_2 = 240$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 37 Question Id : 6406532125104 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Find the optimal bundle of goods 1 and 2 for this consumer. The optimal bundle will consist of $X_1 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

60

Question Number : 38 Question Id : 6406532125105 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Find the optimal bundle of goods 1 and 2 for this consumer. The optimal bundle will consist of $X_2 =$ _____

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

20